Posterior sampling: why we did it

A fitted estimate is useful. A distribution tells us how uncertain that estimate is.

1 Observe the data

Use student evidence to fit the Bayesian model.

2 Describe uncertainty

The posterior assigns probability to plausible weights and noise levels.

3 Sample and summarize

Draw possible model settings to study associations and grade uncertainty.

Sampling adds precision to our calculation—not more student evidence.

Two ways to sample the same posterior

We used the exact solution as a reference for an iterative sampling method.

Direct posterior sampling

20,000 independent draws

Each draw is a fresh sample

Available because this model has an exact solution

Gibbs MCMC

Four chains of related draws

Repeated random updates

Compared with the exact posterior

Why both? Verify the numerical method and understand posterior uncertainty.

How Gibbs MCMC works

MCMC = Markov chain Monte Carlo: a sequence of random updates that explores a posterior.

Sample the weights

Given the current amount of unexplained grade variation.

Sample the noise

Given the current predictor weights.

Repeat the cycle

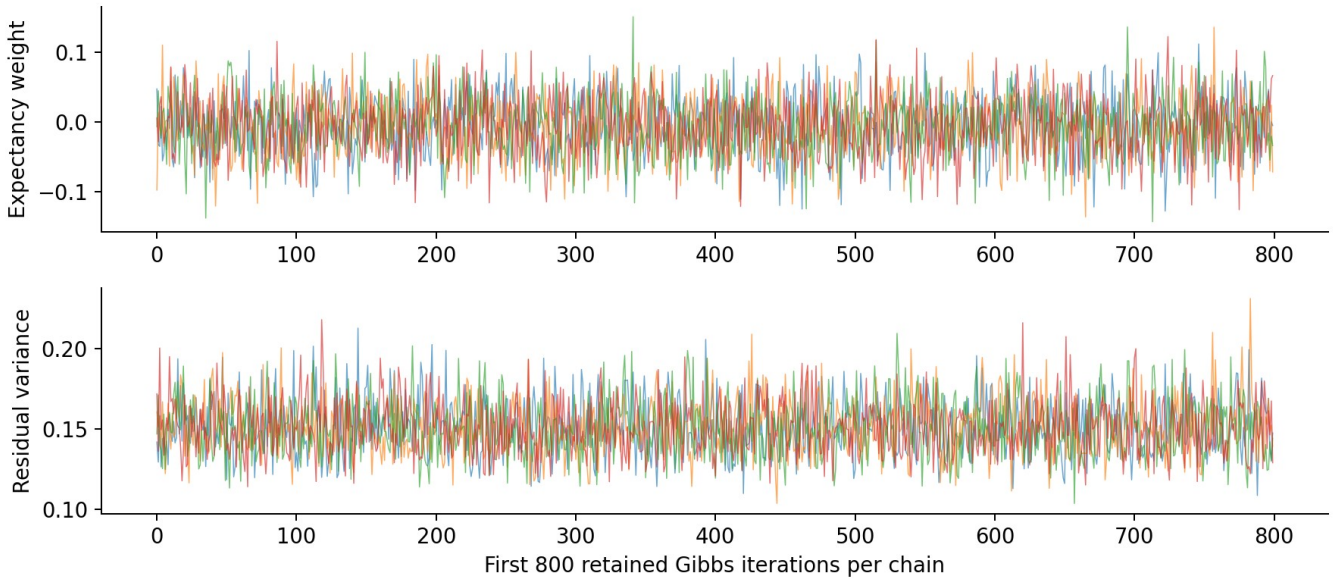
Run four chains from different initial noise levels.

Per chain: discard 1,000 warm-up updates; keep 4,000 draws → 16,000 retained draws.

These draws sample the outcome regression; filled-in predictor values stay fixed.

The sampling agreed with the exact solution

We checked chain behavior, effective information and agreement with the reference answer.



1.0007

Largest R-hat: chains behave similarly

12,194

Smallest bulk effective sample size

≤ 2.01 MCSE

Sampled means vs exact means

The numerical checks support sampling agreement within this fitted model.

What the posterior tells us about the inputs

Earlier study: conditional associations with grades held fixed; 95% credible intervals.

Predictor	Mean points / SD	95% credible interval
LMS page views	+0.34	−0.21 to +0.88
LMS participation	−0.09	−0.63 to +0.44
EVC cost/support	−0.21	−0.84 to +0.41
EVC expectancy	−0.05	−0.70 to +0.60
EVC value	−0.13	−0.73 to +0.47
Wellness	+0.20	−0.39 to +0.78

SD = one standard deviation in the training data. These are associations, not causal effects.

All six intervals include zero: small, uncertain conditional associations.

What this establishes—and what it leaves open

A reliable sampling calculation and a useful forecasting model need different evidence.

Posterior uncertainty	Credible interval: uncertainty about a fitted predictor weight.
Student outcome uncertainty	Prediction interval: uncertainty about one student’s final grade.
Predictive usefulness	Later-student tests: does the model reduce real forecast error?

MCMC checked the calculation. The weekly forecast can still use the exact formula.